

INTE130 – OBJECT-ORIENTED PROGRAMMING

Part (A):

1. One-Person Company Concept:

A sole proprietorship is a company that is established and owned by a single individual, who manages the company and makes all decisions related to its activities. This type of company is characterized by having a legal personality independent of its owner, and the owner's liability is usually limited to the capital invested in the company, which means that the company's obligations do not usually extend to his personal funds. This type of company is suitable for individuals who want to start their own business independently without the need for partners. With technological advances, digital systems have become an important tool for supporting this type of company, providing tools that help the owner organize their business and manage operations such as electronic accounting, sales management, and customer communication, in addition to the use of e-commerce and digital marketing platforms that contribute to expanding the scope of work and reaching a larger number of customers in a more efficient and easier way.

2. Architecture of a One-Person Company:

1. Founder/Owner (Decision Maker)

In a company owned by one person, the founder is primarily responsible for managing the company and making related decisions. He or she determines the project idea and objectives and develops appropriate plans to develop the business and achieve success. They also monitor daily activities and ensure that the business is running as required. Since the company relies on a single individual, the owner is responsible for planning, organizing, and making decisions that affect the future of the company.

2. Digital Agents

Sole proprietorships rely heavily on the use of digital agents or technical tools that help automate certain tasks. These tools include automated customer response software, order management systems, and data analysis tools or digital aids that facilitate business management. Using these tools help business owners save time and effort and make work management more efficient.

3. Business Tasks

Business tasks in an individual company include a set of basic activities that ensure the continuity of commercial activity. The most important of these tasks are marketing products or services, managing sales, communicating with customers, following up on orders, and financial aspects such as preparing invoices and organizing accounts. The owner often takes care of these tasks himself, using digital systems to facilitate their implementation.

3. Role of Multi-Agent Systems:

Multi-agent systems play a significant role in supporting sole proprietors, helping them organize their work and perform various tasks more efficiently. These systems are based on a group of digital agents, with each agent specializing in a specific task within the business. For example, the Marketing Agent works on proposing marketing methods and attracting customers, while the Finance Agent takes care of financial aspects such as tracking profits and expenses and analyzing financial performance. At the same time, the Customer Support Agent communicates with customers, responds to their inquiries, and helps when needed. Although each agent has a different

4. Programming Perspective:

A system for a company managed by one person can be designed using object-oriented programming (OOP) by dividing the system into a set of classes that

represent the basic elements of the business. For example, a Founder Class can be created to represent the owner of the company, who is responsible for creating tasks and making decisions related to business management. An Agent Class can also be created to represent digital agents who perform various tasks or make suggestions to help improve performance. In addition, a Task Class can be used to represent tasks that must be performed within the company, such as marketing, customer service, or data analysis. The Company Class represents the entire company system, connecting the founder, agents, and tasks and organizing how they interact with each other. This approach helps organize the system structure and makes its development and management easier.

Part (B):

1. Business Name:

Online store ERT

2. Business Background:

ERT Store — a Computer Hardware Online Shop Because the project will supply various computer parts and accessories, including keyboards, mice, gaming headsets, etc., for individuals interested in technology or computer games. Store depends on online sales, where buyers can simply find products, they require and choose what to purchase without stepping into a physical store. The project also aims to offer goods of sustainable value.

3. Target Customers:

Among them are university students and people who enjoy home computer games, as well as private individuals using computers for study or work. It also aims to target users who want to purchase a computer accessory, such as a keyboard, mouse, or gaming headset, particularly those who prefer online shopping for its convenience and speed.

4. Why Business Can Be Managed by One Person:

Since most operations are performed online and automatically, one person can run this project. The customers can view the products and place orders on the online

store, where the payment is also made through electronic means without physical presence. Digital agents also assist with logistics, such as arranging orders and processing sales, not to mention dealing with customer inquiries. Hence, the founder can use these digital tools to manage the store more conveniently.

5. Proposed Agents:

Marketing Agent : It promotes the e-store and suggests methods to attract additional customers.

Order Agent : Accountable for tracking and organizing customer requests through the system.

Customer Support Agent : Responsible for responding to customer inquiries and helping as needed

Product Agent : It manages the products in the store, adding new products and updating information related to the items.

6. Example Interaction:

A marketing employee suggests a way to increase sales, proposing to run ads on social media to promote the products. The founder may ask the order processing employees to improve their skills to ensure accuracy and increase productivity, while the Customer support employees provide feedback on customer complaints. This collaboration between the founder and employees helps improve store management, enhance customer service, and make better decisions to grow the business.